

PROCESSSCOPE

Real-Time Operating System Monitoring & Visualization Platform

Version 1.0 Product Design Document

1. Project Overview

Vision

ProcessScope is a web-based operating system monitoring and visualization platform that provides users with a deep understanding of how processes, threads, CPU scheduling, memory allocation, and system resources behave in real time.

Unlike traditional system monitoring tools such as Task Manager or htop, ProcessScope focuses on educational visualization and system analysis by transforming low-level operating system activities into interactive graphical representations.

The platform serves three primary purposes:

- System Monitoring
 - Operating System Education
 - Performance Analysis
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2. Problem Statement

Current monitoring tools provide raw information but fail to explain:

- Why CPU usage spikes
- How processes interact
- How scheduling algorithms work
- Why deadlocks occur
- How memory is allocated
- How resources are shared

ProcessScope bridges the gap between operating system theory and real-world execution.

3. User Types

Computer Science Students

Need to understand:

- Scheduling
- Deadlocks
- Process states
- Resource management

Software Engineers

Need to:

- Monitor applications
- Identify bottlenecks
- Track resource consumption

System Administrators

Need:

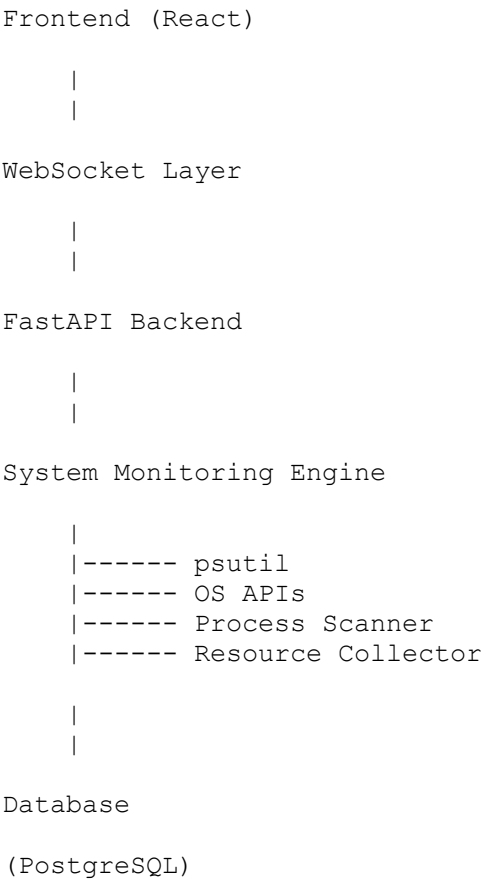
- Real-time monitoring
- Historical analytics
- Process investigations

Educators

Need visual demonstrations of:

- CPU scheduling
- Deadlocks
- Process lifecycle

4. System Architecture



5. Dashboard Module

Purpose

Provide a real-time overview of the operating system.

Components

CPU Overview Card

Displays:

- Total CPU Usage
- CPU Temperature (if available)
- Number of Cores
- Active Threads

Visualization:

CPU
73%



Updates every second.

Memory Overview Card

Displays:

- Total Memory
- Used Memory
- Available Memory

Visualization:

16 GB

Used:
10.3 GB

Free:
5.7 GB

Storage Overview Card

Displays:

- Disk Capacity
- Disk Usage
- Read Speed
- Write Speed

Network Overview Card

Displays:

- Upload Speed
 - Download Speed
 - Active Connections
-

6. Process Explorer Module

Purpose

Display all running processes.

Features

Process Table

Columns:

Field	Description
PID	Process ID
Name	Process Name
CPU	CPU Usage
Memory	Memory Usage
Threads	Thread Count
State	Current State

Search Process

Search by:

- Name
- PID

Example:

`chrome`

Returns all Chrome processes.

Sort Processes

Sort by:

- CPU Usage
- Memory Usage
- Thread Count
- Process Name

Process Detail View

Clicking a process opens:

`Chrome.exe`

Displays:

- Parent Process
- Child Processes
- Start Time
- CPU History
- Memory History
- Threads

7. Process Dependency Tree

Purpose

Visualize process hierarchy.

Example

`explorer.exe`

|— `chrome.exe`

| |— `renderer`

```
|   └─ renderer
|   └─ renderer
└─ vscode.exe
```

Features

Interactive Tree

Expand/Collapse nodes.

Parent Process Tracking

Identify process creators.

Child Process Analysis

Understand process spawning behavior.

8. Thread Explorer

Purpose

Analyze thread-level activity.

Features

Display:

- Thread ID
- CPU Usage
- Creation Time
- State

Thread Visualization

Example:

Chrome

56 Threads

Visual graph showing thread activity.

9. CPU Scheduler Simulator

Purpose

Teach CPU scheduling algorithms visually.

Supported Algorithms

FCFS

First Come First Serve

SJF

Shortest Job First

Priority Scheduling

Priority-Based Execution

Round Robin

Time Quantum Based

Multilevel Queue

Advanced Scheduling

User Input

Process

P1 = 5 ms

P2 = 2 ms

P3 = 8 ms

Output

Gantt Chart

| P1 | P2 | P3 |

Metrics

Calculate:

- Waiting Time
 - Turnaround Time
 - Response Time
 - Throughput
-

10. Process State Visualizer

Purpose

Demonstrate process lifecycle.

States:

New

Ready

Running

Waiting

Terminated

Visualization

New

↓

Ready

↓

Running

↓

Waiting

↓

Running

↓

Terminated

Animated transitions.

11. Resource Allocation Module

Purpose

Visualize resource ownership.

Resources

Examples:

CPU

Printer

Disk

Network

Visualization

P1 → CPU
P2 → Disk
P3 → Memory

Interactive Graph.

12. Deadlock Detection Module

Purpose

Demonstrate deadlock formation.

User Creates Scenario

P1 holds R1
P2 holds R2
P1 requests R2
P2 requests R1

Output

DEADLOCK DETECTED

Features

Resource Allocation Graph

Visual network graph.

Deadlock Explanation

Step-by-step explanation.

Deadlock Resolution Suggestions

Show recovery methods.

13. Historical Analytics

Purpose

Store system activity.

Data Collected

- CPU Usage
 - Memory Usage
 - Process Count
 - Thread Count
 - Network Usage
-

Time Ranges

Last 5 Minutes

Last 1 Hour

Last 24 Hours

Last 7 Days

Charts

- Line Charts
 - Area Charts
 - Heat Maps
-

14. System Replay Engine

Purpose

Replay historical system activity.

Example

2:10 PM

Chrome Started

2:12 PM

CPU Usage Increased

2:13 PM

VS Code Opened

Playback Controls

- Play
 - Pause
 - Fast Forward
 - Rewind
-

This feature acts like:

CCTV footage for operating system activity.

15. Alerts & Anomaly Detection

Purpose

Notify users of unusual behavior.

Rules

High CPU Usage

CPU > 90%

High Memory Usage

RAM > 85%

Excessive Process Creation

100+ processes within 1 minute

Alert Types

- Warning
 - Critical
-

16. AI Performance Analyst

Purpose

Explain system behavior in plain English.

Example

Input:

CPU Spike

AI Output:

Chrome consumed 68% CPU.

Most likely caused by video rendering
or JavaScript execution.

Features

Performance Summary

Daily system report.

Root Cause Suggestions

Identify bottlenecks.

Optimization Tips

Recommend improvements.

17. Technology Stack

Frontend

React
TypeScript
TailwindCSS
React Flow
Framer Motion
Recharts

Backend

FastAPI
Python
psutil
WebSockets

Database

PostgreSQL

Deployment

Frontend:
Vercel

Backend:
Railway
Render
DigitalOcean

18. Future Roadmap

Version 2.0

- Docker Container Monitoring
 - Kubernetes Monitoring
 - Multi-Machine Monitoring
 - Remote Agents
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Version 3.0

- Memory Leak Detection
 - Predictive Performance Analysis
 - AI Resource Optimization
 - Distributed System Monitoring
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Portfolio Description

ProcessScope is a real-time operating system monitoring and visualization platform that transforms low-level system activity into interactive visual insights. The platform provides process tracking, thread analysis, CPU scheduling simulation, deadlock detection, resource allocation graphs, historical replay, anomaly detection, and AI-powered performance analysis. Built using React, TypeScript, FastAPI, WebSockets, PostgreSQL, and system-level monitoring APIs, ProcessScope bridges the gap between operating system theory and practical system analysis.